Geomagnetic positioning without live satellite signals: what is measured and what is calculated

Ankur Srivastava · Version 1.0 · June 2026

Abstract

Geomagnetic positioning uses measurements of Earth's magnetic field to calculate position information for an application. Understanding that process requires a distinction between the field recorded by a sensor, the position produced by the arrangement and the independent reference used to assess it. NSATS's own research and development (R&D) demonstration arrangement records magnetic measurements from eight sensors and uses NSATS's proprietary noise filtering before calculating position. Its records express positions as metres north and east of a known start point. For an organisation considering an implementation, the relevant questions concern the operating environment, the platform carrying the sensors, the application receiving the position and the evidence needed for acceptance. This paper explains the measurement vocabulary and the responsibilities that connect those questions. Scientific field models provide background to the magnetic environment; the licensee's own evaluation establishes what its implementation can do under its intended conditions.

The executive question

A positioning requirement begins with the information an organisation needs to use. An inspection record needs a position associated with the observation being recorded. An operator display needs location information appropriate to the equipment's current task. In either case, the decision concerns the complete path from sensing to an action or record that the application can use.

That requirement can extend to circumstances where satellite positioning is unavailable, degraded or untrusted. Indoor, underground, underwater and dense structural settings raise questions about reception; interference raises questions about whether the received information can be relied upon. These conditions define the problem to be examined. They do not, by themselves, specify what another positioning implementation must deliver.

NSATS's geomagnetic technology determines location using Earth's magnetic field without reliance on live GPS satellite signals. "Live" refers to broadcasts received during positioning. The distinction matters when describing dependencies: information used to establish a starting reference, information used during operation and information used only to assess an output have different roles.

An organisation should specify which task must continue under the intended conditions and what that task needs to know about position. The specification should also identify the platform, its movement and how the application handles an output that cannot support the intended action. A positioning component supplies information; the surrounding application determines how that information is displayed, retained or acted upon.

The operating environments named by NSATS are intended application directions. A licensee still needs to connect its chosen environment to an installation and an assessment. “Indoor” describes a class of setting, while the mounting, surrounding structures and electrical equipment describe the conditions encountered by a particular device. A useful requirement records both.

This paper follows that distinction from the physical signal to the evaluation record. The field measurement, calculated position and assessment reference remain separate throughout. That separation lets a reviewer identify what was observed, what was produced by processing and what information supports the acceptance decision.

Positioning from Earth’s magnetic field · Positioning without GPS satellite signals

Earth’s magnetic field as a signal

Earth’s magnetic field has a large-scale structure, with contributions from different physical sources. The main field originates in the conducting fluid of the outer core. Magnetised material in the crust contributes additional spatial variation, while electrical currents in the upper atmosphere and surrounding space produce disturbances that vary with time. The measured field contains these contributions together. 1

Two public models describe the large-scale main field. The World Magnetic Model (WMM) represents the main field and its slow change in time. The International Geomagnetic Reference Field (IGRF) is a standard mathematical description of the main field, maintained by the International Association of Geomagnetism and Aeronomy. Both provide a common scientific reference for the field at the scale of the planet. Their existence does not mean that a reading beside a structure or operating machine must equal a model value. 1 2

The broad structure is often described approximately as a dipole, but the field at a measurement point is not determined by that approximation alone. Local geology contributes magnetic variation. Buildings, nearby ferrous objects and electrical activity can also affect what a sensor records. A main-field model and a measurement beside equipment therefore describe different levels of detail. 3

Slow change in the main field is known as secular variation. A description of the field consequently needs to distinguish variation with position from variation over time. Moving the sensor through a setting changes where it measures; remaining in the same setting does not make the surrounding field permanently constant. 1

Shorter-term disturbances have several sources. Machinery and changing electrical loads affect the local measurement environment described on NSATS’s operating-environments page. Space weather provides a different source of variation: changes in the interaction between the solar wind and Earth’s magnetosphere alter electrical currents and the magnetic disturbance experienced nearer the surface. These are physical conditions to recognise when describing an assessment, rather than a numerical correction prescribed by this paper. 4

For an implementation review, these distinctions determine what belongs in the environmental description. The record should identify the platform, mounting and relevant surrounding activity. Fixed structures, moving equipment and changes in electrical activity can be recorded separately, so the reviewer can distinguish a change in the installation from a change in the conditions around it. This follows the site's evaluation framework and does not prescribe how the positioning calculation treats those conditions.

For technical readers: the WMM and the IGRF are cited here as scientific background to the field and its variation. They are not a description of NSATS's position-calculation method or evidence for the performance of an implementation. Their published scope and limitations concern the models themselves. The reference position used in an evaluation is a separate record of where the assessed object was.

Underground, on the ground and in the air

What a sensor measures

A magnetic sensor measures the field at the sensor. Field magnitude describes its strength; components describe the field along specified axes. Magnetic field strengths and components are commonly expressed in nanotesla, abbreviated nT. Magnitude and direction are distinct aspects of a vector field. Recording a magnitude does not make it a position, and a direction measurement does not identify where an object is. 5

NSATS's own R&D demonstration arrangement uses eight sensors. Its record format identifies the sensor associated with each observation and provides fields for the raw and filtered magnetic magnitudes. The raw field includes contributions from the platform and environment. NSATS's proprietary noise filtering produces the filtered values from which position is calculated. The public description stops at those inputs and outputs.

The distinction between raw and filtered values concerns their place in the processing sequence. The raw value is the sensor's recorded field before filtering; the filtered value is the corresponding value afterwards. Both remain magnetic-field quantities. The calculated position has a different meaning and unit. A reader can therefore follow the record without treating a change in a field value as a position result.

The sensor identifier also has a practical role in reading a run. It connects the field observations to the derived output recorded for that sensor. An assessment should preserve that association instead of presenting a position trace without identifying which observations and configuration it concerns. A collection of sensor records describes the arrangement used for the assessment; it does not prescribe the hardware of every licensed implementation.

The physical installation is part of this context. The licensee's guidance records the sensor arrangement and mounting, including orientation and distance from ferrous structures and power equipment. It also records the processing unit, power supply and software version. These details identify the configuration to which the observations belong.

Calibration depends on the licensee's sensor. Where the manufacturer provides a procedure, that procedure applies. The installation record states whether calibration was performed and records the magnetic-environment check made before positioning use. A change to the sensor, mounting or

surrounding structure is recorded as a new configuration. This allows an assessment to refer to the equipment actually used without assuming that a previous configuration remains applicable.

For technical readers: NSATS's records may hold field components instead of a magnitude. The record must identify which quantity is present, so a component is not read as a total field magnitude. It also distinguishes a position derived per sensor from a shared derived position repeated across sensor rows. Repeated entries do not establish separate calculated tracks; the record of the arrangement determines how they are interpreted.

What is calculated

NSATS's records express the calculated position relative to a known start point: north and east distances in metres from that start. Reference positions and derived positions use the same origin. This gives the evaluator a common frame in which to read the position output alongside the independently obtained reference.

Relative positions suit that record because they retain the relationship between the assessed positions without disclosing the absolute location. The start point supplies the common origin; each later entry describes a position in relation to it. Publishing positions in this form, with the absolute location withheld, is a publication convention, not a statement about the proprietary position-calculation method.

An implementation must establish how its known start point relates to the application's working information. The licensee's plan identifies the source of that starting reference, how it is supplied during setup and which part of the application uses it. If the application relates an output to an inspection record or operator display, that association belongs in the integration description. This paper does not prescribe a starting-reference instrument.

The setup source should be distinguished from a reference used only during evaluation. A reference receiver or other reference system can supply assessment information without participating in the calculation under examination. Conversely, information that enters the calculation is an operating input and should be identified as such. The evaluation record needs that distinction before an independence claim can be understood.

A suitable reference concerns the same object and observation event as the position being assessed. If the object moves between observations, the assessment must account for that when pairing the records. The reference also has its own uncertainty and conditions of use. Agreement with it is interpreted in that context; the reference is not treated as an unexplained certainty.

The application's use of the output adds another requirement. A position retained from an earlier observation and a newly received position can have different relevance to a moving device. The integration description should identify how observations are associated with their task and how the application handles an interruption or an output it cannot use. These are requirements for the implementation's behaviour, not results reported for NSATS's arrangement.

For technical readers: the record uses a point index to keep observations in sequence and elapsed seconds to relate them to the start. The sensor identifier associates an observation with its source within the arrangement. Those fields support the reading of a record without exposing the absolute location. They do not describe how magnetic measurements are resolved into position.

How positioning is evaluated

Vocabulary for reading an evaluation

The vocabulary separates the observation, the calculated output and the conditions under which both were recorded. A run is the sequence described by one run record. Its movement case describes the movement being assessed, while its setting describes the surrounding environment. The location code identifies the start point without giving the reader its absolute location.

The table lists each term with the identifier used in NSATS's run records and its definition. Movement and setting entries are vocabulary examples only; they report no assessment or outcome.

Term	Record field	Definition
Reference position	`ref_north_m`, `ref_east_m`	Reference position relative to the run's start point (point 0), metres north and east
Derived position	`der_north_m`, `der_east_m`	Derived position relative to the same start point
Start point	`point_index` 0	The first point of the run (point 0): the origin from which reference and derived positions are measured, identified only by its location code
Step	`step_m`	Horizontal distance the reference moved since the previous point
Deviation	`dev_m`, `dev_pct`	At each point the arrangement calculates a position afresh. Deviation is the horizontal distance between the calculated position and the reference position at that point. As a percentage, it is that distance divided by the distance moved since the previous point, multiplied by 100. The start point has no previous point and is excluded.
Run	`run_id`	One recorded sequence of points from a start point; `G-NNN`, sequential (no year, so the ID carries no date)
Movement case	`movement_case`	The movement being assessed, in words: `walking`, `stationary`
Setting	`setting`	The surrounding environment, coarse only, no place names
Location code	`gps_loc_code`	NSATS identifies run locations by code and does not publish coordinates

Table: Vocabulary and record fields; definitions and labels only, with no run data.

In reading an evaluation, keep the vocabulary attached to the specific record. The setting describes where the work falls within a class of environment. The movement case identifies the activity represented. Neither label alone supplies a result for another configuration. The record's reference, installation and assessment scope are needed to interpret its contents.

Published records use a start time and elapsed seconds. A series is identified by its run sequence rather than by run dates. The public location code does not disclose the start-point coordinates. These conventions preserve the sequence and measurement context while withholding absolute locations.

The Validation results and Geomagnetism demonstration records pages are where recorded runs of NSATS's own R&D demonstration arrangement appear, each once it has been approved. A paper defining the record vocabulary does not supply a measured result. Acceptance of a licensee's implementation rests on the assessment of its own intended task.

Applications and licensed intellectual property

Geomagnetic position information can serve an inspection record, an operator display or another function in a licensee's system. The application determines what information is useful, when it is needed and how it is presented or retained. Those requirements connect the licensed positioning function to the surrounding device and its users.

NSATS licenses its intellectual property: patents, designs and copyrighted software. Where the licensed IP takes the form of software, NSATS supplies it under the licence, with its documentation and updates. Each licensee designs and builds its own solution.

Geomagnetism is currently offered under NSATS's regular software licence agreement; further licence forms follow further validation.

The licensee's work includes the sensor hardware, mounting, platform integration and user application. Its installation guidance identifies the configuration, the set-up sequence and the check that confirms position output. Its assessment then examines whether that output meets the task's requirements under the conditions selected for evaluation. Confirming that an installed arrangement produces an output and accepting that output for a particular use are separate decisions.

Integration also connects responsibilities. The part producing position information, the part displaying it and the part retaining a record may be designed or assessed for different purposes. The implementation plan should identify the information passed between them and which component determines the application's response. This keeps the acceptance question tied to the complete task.

Security belongs in that plan. The site's framework addresses unauthorised access to position information, disruption of the positioning task, deliberate magnetic interference, spoofed reference signals and tampering with sensors or processing. The assessment identifies the implementation version, threat model, conditions and assessment owner. Position-validation records do not replace that security assessment.

Geomagnetism IP licensing · Installation and calibration guidance · Security and positioning resilience · Security evaluation

The executive decision

The implementation plan should make the intended use and the means of assessing it explicit. The following questions connect the measurement distinctions in this paper to that plan.

- **Operating environments:** Which environments must the implementation serve, and what structures, equipment and changing activity should its evaluation represent?
- **Platform speeds:** What movement, speeds and stationary stages does the platform's task involve, and how will the observation schedule represent them?
- **Reference sources:** Which independent reference sources are available for the organisation's own evaluation, what uncertainty do they carry, and how will observations be paired with the calculated positions?
- **Integration constraints:** How will the organisation establish the starting reference, mount its chosen sensors and calibrate them where the sensor requires it, supply power and processing, and connect position information to the user application?
- **Security requirements:** What information and actions need protection, which threats belong in the assessment, and what behaviour is required when the application cannot use the position output?
- **Assessment ownership:** Who defines the acceptance criteria, records the configuration and conditions, assesses the evidence and decides whether the implementation meets the intended task?

Scientific background references

These sources explain the magnetic field, its measurement and its variation. They are scientific background, not evidence of NSATS's performance or a description of its proprietary calculation.

1. NOAA National Centers for Environmental Information. *World Magnetic Model (WMM)*. ncei.noaa.gov/products/world-magnetic-model.
2. International Association of Geomagnetism and Aeronomy, hosted by NOAA NCEI. *International Geomagnetic Reference Field (IGRF)*. ncei.noaa.gov/products/international-geomagnetic-reference-field.
3. Lowes, F. J., for IAGA Working Group V-MOD. *IGRF Health Warning, Errors, and Limitations*. ncei.noaa.gov/products/international-geomagnetic-reference-field/health-warning.
4. NOAA / NWS Space Weather Prediction Center. *Geomagnetic Storms*. spaceweather.gov/phenomena/geomagnetic-storms.
5. Natural Resources Canada. *Magnetic components*. geomag.nrcan.gc.ca/mag_fld/comp-en.php.